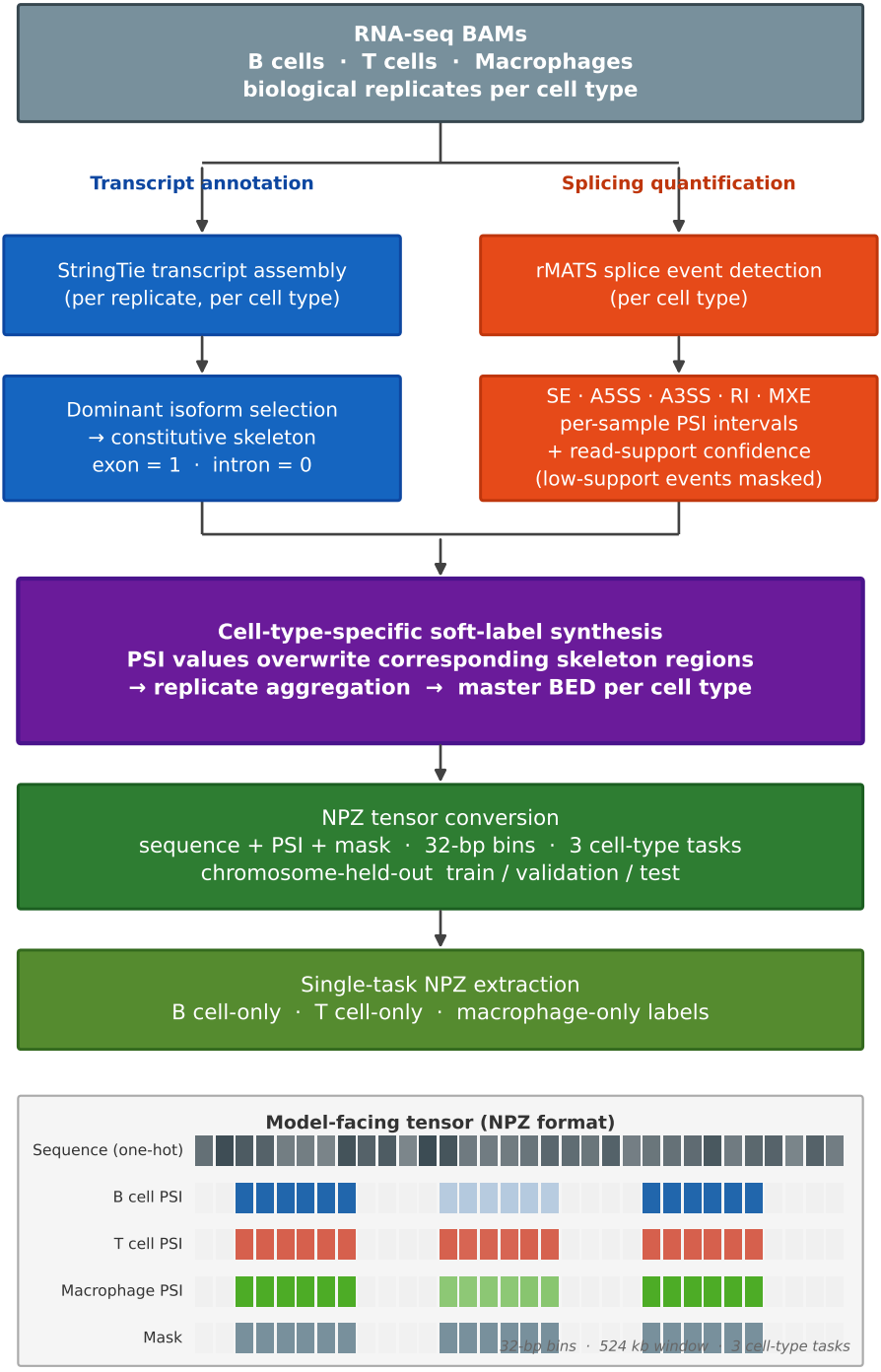
RNA-seq BAMs
B cells · T cells · Macrophages
biological replicates per cell type
Transcript annotation
Splicing quantification
StringTie transcript assembly
(per replicate, per cell type)
rMATS splice event detection
(per cell type)
Dominant isoform selection
→ constitutive skeleton
exon = 1 · intron = 0
SE · A5SS · A3SS · RI · MXE
per-sample PSI intervals
+ read-support confidence
(low-support events masked)
Cell-type-specific soft-label synthesis
PSI values overwrite corresponding skeleton regions
→ replicate aggregation → master BED per cell type
NPZ tensor conversion
sequence + PSI + mask · 32-bp bins · 3 cell-type tasks
chromosome-held-out train / validation / test
Single-task NPZ extraction
B cell-only · T cell-only · macrophage-only labels
Model-facing tensor (NPZ format)
Sequence (one-hot)
B cell PSI
T cell PSI
Macrophage PSI
Mask
32-bp bins · 524 kb window · 3 cell-type tasks